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GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int fd[2];
    pid_t pid1, pid2;

    if (pipe(fd) == -1) {
        perror("pipe failed");
        exit(1);
    }

    pid1 = fork();

    if (pid1 < 0) {
        perror("fork failed");
        exit(1);
    }

    if (pid1 == 0) {
        // Child 1: executes ls -l
        close(fd[0]); // close read end
        dup2(fd[1], STDOUT_FILENO); // redirect stdout to pipe write end
        close(fd[1]);

        execlp("ls", "ls", "-l", NULL);
        perror("exec ls failed");
        exit(1);
    }

    pid2 = fork();

    if (pid2 < 0) {
        perror("fork failed");
        exit(1);
    }

    if (pid2 == 0) {
        // Child 2: executes grep ".c"
        close(fd[1]); // close write end
        dup2(fd[0], STDIN_FILENO); // redirect stdin to pipe read end
        close(fd[0]);

        execlp("grep", "grep", ".c", NULL);
        perror("exec grep failed");
        exit(1);
    }

    // Parent: close both ends and wait for children
    close(fd[0]);
    close(fd[1]);
}

```

```

[ubuntu@ubuntu:~/OSLab/Practical]$ gcc program5.c -o program5
[ubuntu@ubuntu:~/OSLab/Practical]$ ./program5
-rw-rw-r-- 1 ubuntu ubuntu 741 Sep 10 10:41 program2.c
-rw-rw-r-- 1 ubuntu ubuntu 1077 Sep 10 10:51 program3.c
-rw-rw-r-- 1 ubuntu ubuntu 928 Sep 10 11:01 program5.c
-rw-rw-r-- 1 ubuntu ubuntu 908 Aug 25 11:54 program_1.c
-rw-rw-r-- 1 ubuntu ubuntu 1261 Aug 6 22:33 program_2.c
-rw-rw-r-- 1 ubuntu ubuntu 46 Aug 6 22:40 source
ubuntu@ubuntu:~/OSLab/Practical$

```

Output